

Bridge Day 3 INSTRUCTOR KEY

Types, Casting, and Truthiness

Learning sequence: Predict - Trace - Explain - Test - Interpret

Instructor key with answers in student response locations

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Concrete engineering situation

Your team is continuing the pump-flow record from Day 2. Today the values arrive from an intake form before they are used in a notebook. Some entries arrive as text, some are already numbers, and some fields are blank. Before the team can make safe threshold decisions tomorrow, the record has to distinguish type, conversion, and truthiness.

The problem today is not branch routing. The problem is whether a value is text, a number, or Boolean-like evidence, and whether a conversion or truthiness check is safe to trust.

Engineering task	Prepare a pump-flow intake record so later threshold and go/no-go decisions are based on converted values rather than misleading text labels.
Computational move	Classify values by type, cast text when safe, and interpret truthiness without confusing a non-empty label for an approved result.
Python focus	<code>str</code> , <code>float</code> , <code>int</code> , <code>bool</code> , <code>type(...)</code> , <code>float(...)</code> , empty string, non-empty string, zero, and nonzero values.
Evidence to keep	original value, Python type, cast expression, cast result, truthiness result, and one corrected intake-note sentence.

Day 3 type and truthiness rules

1. Text that looks like a number is still text until code converts it. Example: "18.6" is a string, while 18.6 is a float.
2. A cast such as `float("18.6")` produces a numeric value, but it does not change the original text variable unless an assignment stores the result.
3. Empty strings are falsey. Non-empty strings are truthy, even when the text says "no" or "0".
4. Numeric zero is falsey. Most nonzero numbers are truthy.
5. Truthiness is evidence about empty/non-empty or zero/nonzero status; it is not the same as engineering approval.

1. Read the intake record as typed data

recognize

predict

Use the intake record below. First identify what kind of value each line stores. Do not make a go/no-go decision yet.

```
# Intake form values from the pump station
pump_id = "P-22"
target_flow_text = "20.0"
measured_flow_text = "18.6"
operator_note = "needs review"
approved_text = "no"
spare_comment = ""
retry_count = 0
sensor_ready = True

# Safe conversions for later calculations
target_flow_lpm = float(target_flow_text)
measured_flow_lpm = float(measured_flow_text)
gap_lpm = target_flow_lpm - measured_flow_lpm
```

Pts.	Prompt	My evidence
1	Which two variables store numbers as text before conversion?	Key: <code>target_flow_text</code> and <code>measured_flow_text</code> .
1	Which variable stores an empty string?	Key: <code>spare_comment</code> stores "".
1	Which variable stores a whole-number count rather than text?	Key: <code>retry_count</code> .
1	Which variable already stores a Boolean value?	Key: <code>sensor_ready</code> stores <code>True</code> .

2. Classify type before using a value

recognize

explain

Complete the type table. Use Python type names such as `str`, `int`, `float`, and `bool`. Then write why the type matters for later work.

Pts.	Variable or expression	Type	Why this matters before a threshold check
2	<code>target_flow_text</code>	Key: <code>str</code>	Key: Original text "20.0"; not safe as a numeric threshold until cast.
2	<code>target_flow_lpm</code>	Key: <code>float</code>	Key: Converted numeric value 20.0 for calculations.
2	<code>approved_text</code>	Key: <code>str</code>	Key: Stores label "no"; truthiness is not approval.
2	<code>sensor_ready</code>	Key: <code>bool</code>	Key: Already stores Boolean evidence.

Common trap to check

The text "20.0" may look like a number, but Python will treat it as text until it is converted. A reliable engineering record names both the original field and the converted field.

3. Cast text to numbers without rewriting the source record**predict****trace**

A cast produces a new value. It changes a stored variable only if the cast result is assigned to a name. Complete the table.

Pts.	Expression or assignment	Value produced or stored	What changed?
2	<code>float(target_flow_text)</code>	Key: 20.0	Key: Produces a float; original text variable is unchanged unless assigned.
2	<code>float(measured_flow_text)</code>	Key: 18.6	Key: Produces a float from the measured-flow text.
2	<code>target_flow_lpm = float(target_flow_text)</code>	Key: 20.0	Key: Stores the converted float in a new numeric field.
2	<code>gap_lpm = target_flow_lpm - measured_flow_lpm</code>	Key: 1.4	Key: Uses converted numeric fields: 20.0 - 18.6.

Pts.	Prompt	My response
2	Why should the notebook keep the original text field and the converted numeric field separate?	Key: The text field preserves the source record; the numeric field is safe for calculations. Keeping both supports audit and debugging.
2	What would be unsafe about calculating with <code>target_flow_text</code> as if it were already a number?	Key: It is a string, so arithmetic may fail or act like text rather than numeric calculation. Cast first.

4. Interpret truthiness without mistaking labels for approval**predict****explain**

Python truthiness can be useful, but it can also mislead an engineering record when text labels are treated like decisions. Complete the table.

Pts.	Value or variable	Truthy or falsey?	Engineering interpretation
1	spare_comment storing ""	Key: falsey	Key: Empty string; no comment text.
1	operator_note storing "needs review"	Key: truthy	Key: Non-empty string; still a review note, not approval.
1	approved_text storing "no"	Key: truthy	Key: Non-empty string, but text means not approved.
1	retry_count storing 0	Key: falsey	Key: Numeric zero is falsey.
1	sensor_ready storing True	Key: truthy	Key: Actual Boolean True.

Truthiness warning

`bool("no")` is True because the string is not empty. That does not mean the operator approved the record.

Pts.	Prompt	My response
2	In one sentence, explain why truthiness is not the same as engineering approval.	Key: A non-empty string such as "no" is truthy, but the actual text still means not approved.

5. Debug a type or truthiness mistake[debug](#)[test](#)[explain](#)

A teammate writes this intake note before tomorrow's threshold check:

Intake note to debug

The record is approved because `approved_text` has a value, and the flow gap can be calculated directly from the two text fields.

Pts.	Prompt	My response
2	What part of the teammate's note confuses truthiness with approval?	Key: It treats <code>approved_text</code> having any non-empty value as approval, even though the value is <code>no</code> .
2	What part of the note confuses text values with numeric values?	Key: It says the flow gap can be calculated directly from the text fields; they should be cast to floats first.
2	Write one non-mutating Python expression that checks the type of <code>approved_text</code> .	Key: <code>type(approved_text)</code> .
2	Rewrite the intake note so it separates text fields, converted numeric fields, and actual approval evidence.	Key: The intake stores text fields such as <code>approved_text = "no"</code> and numeric fields such as <code>target_flow_lpm = 20.0</code> after casting. Approval should be checked explicitly from the approval value/rule, not from truthiness alone.

Bridge to Day 4

Tomorrow's boundary checks will use comparison operators such as `<`, `<=`, and `==`. Those checks are safer when today's type evidence is already clear: converted numbers for numeric thresholds, and explicit labels or Boolean values for decision evidence.

6. Mastery check and support next step**transfer****explain**

Use your own evidence from this workbooklet. Do not write a generic answer.

Pts.	Prompt	My response
2	Give one example where text looked like a number but needed a cast before safe calculation.	Key: "20.0" in <code>target_flow_text</code> must be cast with <code>float(...)</code> before numeric threshold work.
2	Give one example where truthiness could mislead an engineering record if the team did not read the actual value.	Key: <code>bool("no")</code> is <code>True</code> because the string is non-empty, but the label says no.

My mastery check

Mark the strongest statement that is true after this workbooklet.

☐	Secure: I can classify a value's type, predict a safe cast, and explain why truthiness is not the same as approval.
☐	Developing: I can identify some types or casts, but I still need support with string truthiness or converted fields.
☐	Needs support: I need a smaller example before I can use type and truthiness evidence in a record.

My next support move

My issue is mostly: setup / Python syntax / tracing / type conversion / truthiness / confidence / time.

The first value where I got uncertain was: _____

My next small action is: ask a partner / ask instructor / rebuild the type table / rerun one expression / try the recovery version.

Bridge Day 3 INSTRUCTOR KEY

VIP Evidence Handoff

Learning sequence: Predict - Trace - Explain - Test - Interpret

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Why engineers use this page

Revisit the same calculation notebook through type evidence: preserve raw input, convert numeric text, calculate, and format output before independent work.

Decision boundary: Code is useful only when its stored state, visible result, assumptions, units, limits, and tests support a defensible engineering interpretation.

Construct readiness before independent work

New authoring tools: numeric literal, arithmetic expression, floor division, modulo, exponentiation, input call, int conversion, float conversion, f string, import statement, math library call

Carried authoring tools: string literal, assignment, print call, sequential execution, exact output

Supplied-code exposure only: none

An exposure-only construct may be traced in complete supplied code, but the independent task will not require students to invent it before its authoring day.

Notebook cells and task constructs

Activity	Work	Stable cells	Required authoring constructs
18.3	Minutes Conversion	18-3-work / 18-3-transfer / 18-3-cold	arithmetic expression
18.4	Square Root Check	18-4-work / 18-4-transfer / 18-4-cold	f string
18.5	Kit Count Plan	18-5-work / 18-5-transfer / 18-5-cold	profile-level contract
18.6	Rectangle Diagonal	18-6-work / 18-6-transfer / 18-6-cold	f string
18.7	Grid Path Distance	18-7-work / 18-7-transfer / 18-7-cold	f string, input call, int conversion

Readiness evidence

- Write one example showing the raw text returned by input, the converted numeric value, and the calculation that becomes legal after conversion.

Model response: A correct example distinguishes a string such as "8" from int value 8 or a decimal string from its float, then names arithmetic that uses the converted value.

- For one Studio problem, name the required numeric type, output precision, and a test that would expose a missing or incorrect conversion.

Model response: Credit an activity-specific type and precision plus a diagnostic input; the answer must explain what failure or wrong output would reveal the conversion defect.

Timing and help trigger

Record a first prediction before running. If you still lack an executable plan after about 8 focused minutes, use the linked ThinkCSPy refresher or the supplied model. If two attempts fail for the same named requirement, write the exact mismatch and ask a peer mentor or instructor a targeted question. Timing is diagnostic evidence, not a speed grade. **Start or elapsed minutes:** _____

My help trigger: _____

Engineering Evidence Record

Complete one record from a model, transfer, or Independent Program Studio build. Generic statements are not sufficient; use actual values, a real revision, and one concrete next test.

Evidence field	Record
Prediction	A specific expected state or output before execution.
Observed Result	The actual state or output, including a value or exact text.
Revision Reason	The defect or mismatch and the change that addresses it.
Engineering Meaning	How the evidence changes or supports the engineering decision.
Units Or Not Applicable	The physical unit, or the evidence type when no unit applies.
Assumption	One condition accepted as true for this model or rule.
Model Limit	One situation the result does not justify or predict.
Next Test	A concrete boundary, invalid, or alternate case with an expected result.
Help Trigger	The exact unresolved point that would trigger a targeted question.
Minutes Used	Approximate minutes from first prediction through revision.

Notebook and submission procedure

1. Attempt, predict, run, inspect, and revise before opening a reveal.
2. Complete the end-of-notebook Independent Program Studio without a reveal or copied solution. Only those visibly labeled Studio cells are scored for independent mastery.
3. Save or download the completed notebook as one `.ipynb` file.
4. Upload that one notebook to the matching Gradescope assignment. Do not export a `.py` file or create a ZIP.